

Design of multimedia services

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This paper addresses the fundamentals of multimedia service design. It draws on contemporary experience with the design and delivery of large-scale multimedia services to present a comprehensive treatment of the total design process, from initial ideas through to the detailed design of individual applications. It draws an essential difference between service design, which embodies the commercial rationale, business opportunity and technology exploitation, and application creation which is the design and creation of the applications comprising the service. The critical dependencies of hardware platform, user interface functionality and application styleguide are discussed and essential guidelines provided. Application design is comprehensively covered with attention given to the issues of media design and display, user interface consistency and usability.

1. Introduction

1.1 Initial concepts

Before a 'service' can be designed it must be conceived. There is fundamental commercial and technological research to be done before embarking on the long journey of designing a multimedia service and the applications which live therein. There is an order which must be obeyed to achieve success and a structure to be designed and built before it can be decorated.

High technology projects are often late, over budget and much more complex than first thought. There are many examples in the world to substantiate this statement. This does not have to be the case but how do we ensure success? Though this paper cannot answer all these questions, it attempts to identify and relate the issues and tasks which have to be integrated to create a multimedia service and to populate that service with well-designed and timely applications. Experience with previous multimedia projects is now building and there are many lessons to be learnt.

Technology-based companies like BT often perceive multimedia services as an exercise in applying incremental improvements in technology, building on what has been before. The very small amount of successful multimedia world-wide is generally confined to non-networked personal products such as CD-ROMs. There is no model for the successful design and implementation of networked multimedia services, though this may be remedied through

the many world-wide video on demand (VoD) trials being planned and implemented.

This paper divides the creation of a multimedia service into two parts.

Service design — which embodies the commercial rationale, business opportunity and technology exploitation. It also includes the scope and scale of the service and defines the service hardware and software platform. The outline requirements for the applications must be established in aspirational terms to drive the platform functionality and choice. The platform functional specification and user interface (UI) specification are established at this stage, thus providing a generic environment for later application creation. Application creation is not started during this phase.

Application creation — the design and creation of the applications which comprise the service. A function of an application is to present a user interface to the user which provides navigation and selection of multimedia content. Creative design, usability and content, within the platform technical constraints, are the key features of a multimedia application.

2. Service design

The service design process is described in this section and illustrated in Fig 1.

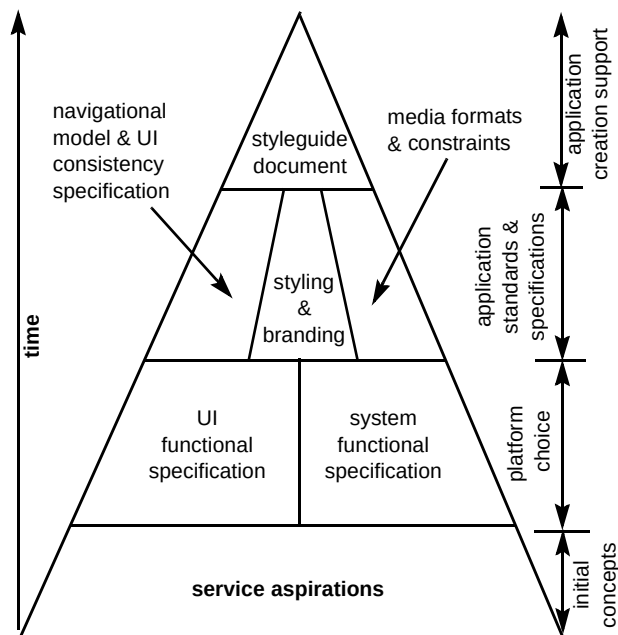


Fig 1 Service design pyramid.

Figure 1 shows the service design process and its elements. It is pertinent to the entire section and the subsections which describe and explain the elements in detail. The time arrow indicates the correct sequence of work. The pyramid shape shows the cumulative build up of knowledge and the interdependency of the elements, cumulating in the production of the styleguide document, which provides the basis for application development.

2.1 Service aspirations

To drive the service design forwards the basic aspirations for the service must be established. There are two components:

- short or target list of business strands to be offered by the service, a business strand being defined as a generic commercial area within the service, e.g. shopping,
- initial concepts of how each business strand can be represented by interactive multimedia technology.

A requirements capture exercise is generally necessary to establish the generic requirements of each business strand. This could be done with a selected 'information provider' partner for each area, who would collaborate in the development of a 'lead' application, and would be offered a place in the service start-up line up. Alternatively a non-aligned source can be sought.

The requirements capture will take the form of a number of workshops whose output will be aspirational specification documents covering:

- typical ranges of products to be offered by each business strand,
- product categories and themes for selection and searching,
- typical total amount of product to be offered,
- the mix of media to provide information about the products, for user navigation and selection,
- choice of media used to represent the product contents — for entertainment services the 'product' is data transmitted to the user, e.g. a movie or TV programme, while for other business strands, such as retailing, catalogue stills and short video clips will be used to provide information about the product,
- business transactions required,
- usage statistics required,
- UI and user interaction.

2.2 Choosing the platform

The aspirations for the service, described in section 2.1, are to some extent preconditioned by a knowledge of the current state of multimedia technology. The UI and user interaction options available are tested and balanced against the aspirations for product representation and display. The ultimate choice of platform will be a design compromise.

The factors for consideration in choosing the platform are:

- display technology — TV, computer monitor, LCD, etc,
- display attributes — size, resolution, visibility, siting and ambient lighting,
- UI interaction device — TV-style remote control unit, touch screen, mouse,
- media storage and distribution — the total on-line media requirement for the service and the distribution of its storage between a central server and local storage device,
- media types — the different media types to be used by the service (video, stills, text, audio), the critical media type being video because of its high storage requirement and impact on network bandwidth,

- performance — the service performance requirements for media display and user interaction feedback will be quantified as maximum response times for all functions within the service, e.g. time to display a video,
- cost — the functionality and performance of the platform are directly related to its cost, the final specification always reflecting a trade-off.
- connection to external systems — requirements for business transaction or networking to external servers.

This paper looks at generic rather than particular multimedia platform solutions and a generalised platform architecture is depicted in Fig 2.

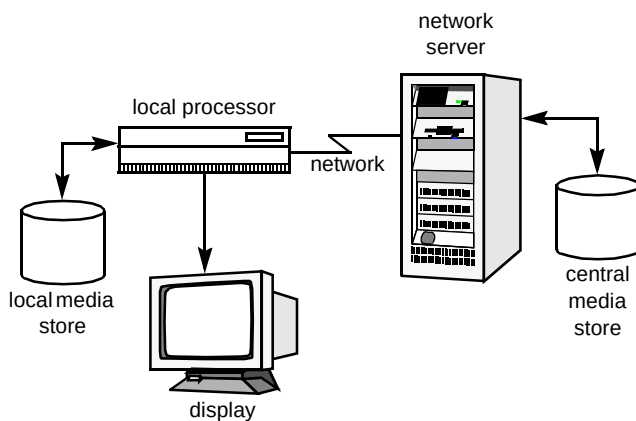


Fig 2 Generic multimedia platform architecture

This generic architectural model is applicable to most particular platform solutions. Though the relative importance of the central components, the network server and the central media store may vary, the model holds good as a starting point for the design of new service platforms.

There are five principal components of the generic multimedia platform.

- Local processor — the local processor executes the user interface code, driving the main display unit and peripherals and providing an interactive capability for user interactions. The local processor provides the local component of the service processing power. The functionality of the local processor is dependent on the distribution of the application media contents between the central and local stores and also the performance criteria for finding and displaying the media content. For some 'stand-alone' kiosk services there is no network connection and a single processor supports the complete service. For others, such as VoD, the local processor provides the UI, calling central services over the network, when required. The UI also provides the

means of displaying central data on the display, for example, playing movies transmitted from the central media store.

- Local media store — the local media store can provide the media content displayed by the application. This comprises still images, text, sound and video. Locally stored video is generally confined to shorts clips because of the high storage requirement. In a kiosk service, typically, there would not be more than 30—60 minutes of stored video. If stored on a hard disk the stored video can be swapped out, by overnight 'store and forward' techniques, from the central media store. With VoD, there is no local media store, all the application content being held on the central media store and supplied to the UI, on demand.
- Central media store — the central media store generally provides a centrally networked source of media content required by the application, comprising still images, text, sound and video. Centrally stored items are generally longer videos and even feature films in the case of VoD services. The central media store can have two functions depending on the service requirements and network bandwidth:
 - to store and forward media to a local processor unit, either 'on demand' or as part of an 'off-line' contents maintenance strategy,
 - to supply media content 'on demand' to local processor units for real time display.
- Network server — the network server provides functions which are executed by a client/server call from a local processor unit. A function could relate to content, for example, finding and transmitting a video clip, or to a business transaction. In the case of a VoD service the central server is all important and can comprise a number of linked devices providing the UI, the application media contents and the business transaction capability in a multi-user, real time environment. For kiosk services the network server provides text database retrieval on demand, for example, retail product prices and availability. It could also provide real time still images if they are stored in a suitably compressed form, such as JPEG. If the network link were broadband, real time uncompressed still images and compressed video could be provided. Broadband provision is unlikely for kiosk services. For this type of application, the media contents could be held in the local media store which would be replenished off-line by the network server.
- Network — network bandwidth is a trade-off between the aspirational media and performance requirements for the service and the installation time-scales and operating costs of network technology. For VoD services the exploitation of broadband technology is an

explicit part of the service concept. For other services the 'ubiquity' of the network is an important factor in the time-scales for exploitation and is intimately connected with the strategy for media distribution and storage, since performance is critical to service usability and user acceptance.

2.3 Platform functional specification

The platform functional specification defines the functional behaviour of the platform. It translates the service requirements into functions which form the application building blocks. The platform functional specification is expressed at a high level but with sufficient detail to enable a software developer to produce a technical specification and write the computer code. The scope of the functions covers:

- product data search and retrieval,
- media retrieval (video, stills, audio),
- business transactions,
- usage statistics,
- gateways to other systems.

The aspirational requirements for the service, as captured and documented through requirements capture workshops, are often not fully supported by the capability of the platform and some downgrading of expectations will occur. Service design has to manage compromise so as to maintain acceptable levels of service viability, despite platform constraints.

2.4 UI functional specification

The UI functional specification defines the building blocks which provide the application designer with the tools to design the application UI. It defines the user interaction paradigms which are the routes to application content searching, navigation and media display.

These interaction paradigms, using the previously defined interaction devices, are specified in detail. The UI specification assigns and maps user action to functionality in a generic, non-application specific manner, setting out generic rules for later incorporation in application design standards. The UI specification covers:

- selection methods for 'product' search and UI interactions to specify search criteria,

- display standards for the media returned following a product search — the mix of media types is defined here and, in particular, the control and access to video material are specified, examples being the standards for continuous video loops and video carousels,
- selection of on-screen objects and feedback,
- UI components of service specific functions, such as 'shopping basket' for retail services and 'movie pay-and-play' for entertainment services.

2.5 The navigation model and UI consistency

The navigation model shapes the generic UI specification one step nearer the applications, providing one or more navigational templates which when tailored for specific applications, provide the core, search, navigation and display engines for those applications. The navigation models are the integration of the components defined by the UI functional specification and are a higher level concept. At this stage, basis navigation conventions are agreed to provide consistent ways implementing basic functionality throughout the service, e.g. a consistent way of exiting all applications. This may simply mean adopting only one possibility, as defined in the UI specification, or designing a more complex structure from UI building blocks.

A common thread of UI consistency, once defined, will be built into all navigation models, to give a consistent core 'feel' throughout the service. The 'look' will and should vary throughout the applications, to reflect the aspirations of the service providers.

The navigation models are essential components of the service design process and are themselves essential components of the styleguide document, described in section 2.8.

2.6 Media specification, formats and constraints

Media specification, formats and constraints are major components of the design of a multimedia service, influencing media contents production, media storage on the system and performance of the service with respect to media retrieval and display.

Some basic media standards are fixed by the platform choice while others will offer some flexibility and choice, allowing final decisions to be made as part of the application development process, discussed in section 3.1. In particular, video standards are critically important, as video is the most resource-hungry of all the media, being exceptionally difficult to produce, encode, store and display, in comparison with the other media types.

A media specification, format and constraints document will be produced at this time. This will progress the standards explicitly and implicitly emanating from the hardware platform and will form the basis of the media section in the styleguide document which will be used by the application development teams. This document will, in general, be expressed in technical terms and will form the definitive technical reference for the service. It will be available to application design teams which will use it for background reference, using the styleguide as their primary reference for design information and support. The document will express media and performance constraints in purely technical terms and will not attempt to develop design guidelines or advice, which are solely the province of the styleguide document. This document is a major input into the styleguide document, described in section 2.8.

2.7 Overall branding and styling

The overall branding and styling represent the fundamental values and character of the service and must be in position as early as possible so that they can be included in the styleguide and hence available to application developers before the commencement of application development.

The ownership of the service and hence the assignment of branding are fundamental decisions to be made. For a simple service with a single service provider using their own brand throughout there is no problem. For more complex services, for example VOD services, with many information providers, the situation is much more complex. In this case the overall service provider must decide ownership of the 'top levels' of the service and the use of their own branding in these areas. The transition to the branding of the information providers will often require negotiation and the information providers will have to 'buy into' the overall standards and values of the service, otherwise it will be difficult to sustain the long-term commercial partnership which is generally desired by both parties. For the above reasons it is imperative that the branding and overall design standards are put in place before the applications are started, otherwise many misunderstandings and mistakes will occur involving loss of time, credibility and expensive reworking of designs and media content.

The place to document overall branding and styling is the styleguide which is described in the next section. However, examples of the 'top-level' design work would generally be available as part of an on-going consultancy exercise and efforts would be made to involve and consult information provider partners. However, it is the service owner who ultimately sets the rules for itself at the top levels of the services and, by negotiation, in other branded areas of the service, i.e. the applications.

2.8 Styleguide document

The application styleguide is arguably the most critical and important document produced by the service design process. It documents the service design standards, as described in sections 2.5 — 2.7, to provide the definitive reference manual for application design and media contents production.

It is important to understand the significance of the main inputs to the styleguide which are repeated below:

- the navigation model and UI consistency,
- media specification, formats and constraints,
- overall branding and styling.

The styleguide interprets, where necessary, the service features and constraints to provide mandatory guidelines and offers design alternatives where possible.

The styleguide document cannot be written unless the three principal inputs exist. It is then possible for an experienced interaction designer to produce the document. However, it is not recommended that this approach is taken. The best approach is to have a fully integrated service design team from the start, involved in all aspects of service design, especially the three direct inputs to the styleguide. The styleguide document will then grow organically from the service design process, being fully available and supportable at the right time.

The styleguide is a 'living' document which is fully supported as part of the application creation process, which is the next major section of this paper.

3. Application design

3.1 Process model

In order to be able to design an application, a process model is required. The following model shows the different activities that take place during the early stages of application creation, up to and including the definition and sign off of the application design document. Requirements capture techniques (see section 3.3) are required to capture the requirements of customers, in early meetings and workshops with the customer. These requirements should not be constrained to the functionality of the technology. Customers should be allowed to express their aspirations for the application and service. These aspirations can be used to enhance the technology, during future upgrades. Included in this process, is the information on corporate styling and

branding. Requirements and understanding of the brand values the customer wishes to express within the application and service will be captured and understood.

This information, and the information included in the styleguide (see section 3.2), allows the creative designer or interaction designer to work out the interaction and navigation of the application. These are drawn up as visual storyboards (see section 3.3). These storyboards are reviewed with the customer at regular workshops. Once agreed these are worked into concept screen designs. These screen designs show how the application would look once complete. They would follow all the consistency requirements of the styleguide. Again, on agreement, these are put together by a prototyper, using the visual storyboards, and a prototyping authoring package and turned into a concept demonstrator.

This demonstrator is tested (see section 3.3) with the target users. Human factors facilitators are able to explore and test the usability of the application with the users. This valuable information is fed back into the design process. Where required, the storyboards and the demonstrator are altered to reflect the user testing recommendations.

During this whole process, information is being gathered for the application design document, sometimes called the design brief. This is the definitive document for the design of the application, and must be agreed and signed by the customer.

All the activities described, and shown in the model (see Fig 3) must be co-ordinated. Therefore an overall application design manager is required at least until completion and sign-off of the application design document.

3.2 Design support

3.2.1 Styleguide

The styleguide is the 'applications bible'. It is used by the application design team, during development of the visual and technical storyboards, during early prototyping and the production of the application design document. Its purpose is fivefold:

- to document the navigation model of the service, from the UI functional specification developed through the service design process, thus ensuring that the user interaction and navigation is consistent in all generic aspects of each application, i.e. so that all applications work in the same way — for example, during payment, the order in which the user is expected to enter data would be consistent, thus reducing the learning task and minimising confusion within the different applications,

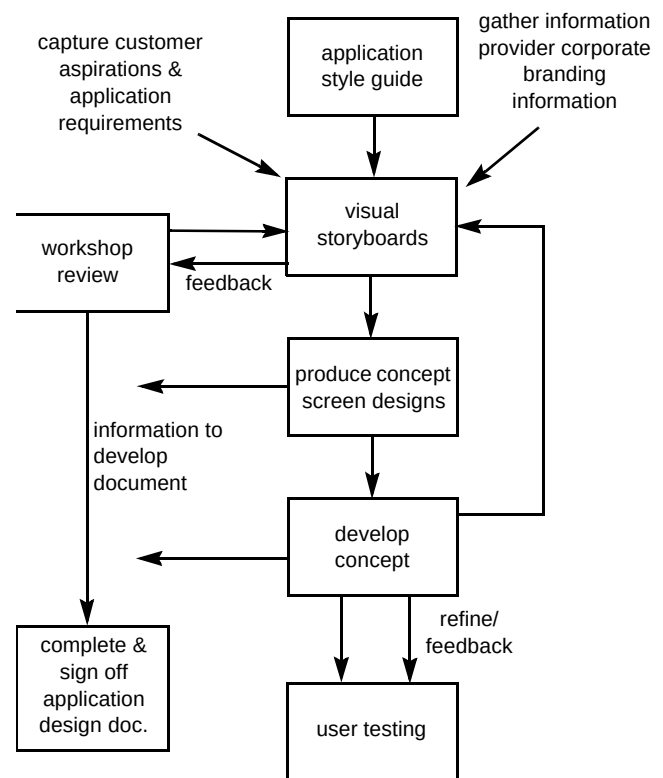


Fig 3 Application design process model.

- to prescribe, in detail, core areas of the user interface consistency, e.g. ensuring that the user can always expect help, information or error messages to be consistent in both look and style — interface consistency, however, should not restrict the styling and branding of a third-party information provider,
- to state mandatory core standards where applicable — these may apply for designing on the target screen displays, or for the platform,
- to document all user interface constraints imposed by the platform,
- to promote best practice, within the 'art of the possible', in user interface design, e.g. the amount of text to be used on screen, and the minimum size of the text that should be on screen,

In addition, the styleguide must cover the following topics in detail:

- Styling and branding

This section of the styleguide needs to clearly document the positioning of the brand and brand values of the service provider. These decisions will have been made during the service design stage, before application design commences. The brand and its positioning will need to be reflected in a branding

system that will be imposed on to the application design. The branding system will reflect the brand of the service provider, the brand of a third-party information provider and the transition between the two.

The branding system would define the attributes that reflect the brand — the use of corporate fonts, the positioning of service and application titles, the position of corporate and third-party brands or logos, the use and positioning of sponsorship logos, and the transition between corporate and third-party corporate branding.

The BT market trial branding system, for example, carries the feel of the BT corporate brand through to the main screen of each application. Each application main menu has a branding strip on the left hand side of the screen. This strip contains the BT brand, the icon that represents the application area and the title of the application.

- **User interface consistency**

The user interface should cover what the user sees on the display screen, the interaction devices used to interact with the screen, and any literature that may accompany the service.

The following are the sections that can be expected to be covered in this area of the styleguide.

User interface display — information on the type of display to be used, is provided. For example a computer screen, a TV screen or a LCD display. To design for a particular display, the designer needs information on the size and resolution of that display. Environmental information is also specified, such as, information on ambient lighting, reflections, and the viewing distance of the user from a screen.

Interaction device — this section gives full details on the device that will be used to interact with the application. This may be a mouse, touchscreen, or infrared remote control. Each device has limitations which affects the design of the user interface. For example, a touchscreen would have a minimum setting for screen objects.

Colour — colour display is dependent on the display hardware being used. For example, a TV in the UK can only truly display PAL friendly colours — colours outside of this range will badly bleed across the screen or flicker. Technical standards are available, and should be quoted. Human Factors standards provide information on how many colours should be used on screen at one time and describe the emotional significance of certain colours, e.g. red is representative of stop or danger.

Text — this section recommends usage of fonts, within an application. These recommendations include text sizes, font types and text presentation. Minimum font sizes are required for different text. This text could be a title, product description or price information. Font sizes are dependent on the viewing distance of the user to the display. Some text will not change through the lifetime of the application. This ‘static’ text is anti-aliased to the background image. The font used is normally a corporate font, or one that is best displayed on the target display. However, some information must be updated regularly, for example, news information. This information is stored in databases, and cannot be anti-aliased, giving it a less smooth look. This ‘dynamic’ text is represented in a default system font, typically Helvetica. Presentation of text is available from human factors standards, and covers the tone of the text or phrase, where it should be displayed, and how it should be displayed.

Sound — sound can be represented in three ways — audio feedback, background music and voice over. This section details the use of music and voice over, especially when it should be used and how frequently. The tone and tempo of the music should reflect the desired atmosphere sought for the application. The music should not be too repetitive, this is annoying to the user. The same applies to ‘voice overs’. An appropriate voice should be chosen for the application. ‘Voice overs’ should support the text used on screen.

Imagery — the way in which imagery is presented on screen is dependent on the display hardware and the size of the display. Recommendations need to be made for background and foreground static images as well as video content.

Selection — selection can be direct or indirect. Direct selection occurs when a single interaction causes an action. For example, a choice on screen is associated with the number ‘1’; by pressing ‘1’ on a remote control the user would automatically be taken to a new screen representing that choice. On interaction, both audio and visual feedback is given. Indirect selection requires two interactions from a user. This first is to move a ‘focus’ visual highlight from one option to another, the second interaction is to select the focused option. Two different visual highlights are required for this method. A user should be able to recognise the different selection method and interact accordingly. If the selection choice is small, the direct method is generally preferred.

Feedback — feedback can be tactile, audible and visual. The user will get tactile feedback as they press a button on a remote control, a visual feedback will be displayed on the screen, and an audio feedback can be heard.

Messaging — this section gives recommendations for global system error and information messages, and local application error and information messages. These recommendations will cover how the messages will be displayed, to draw attention to the user, and the wording associated with the message. Messages are normally represented in pop-up boxes that appear over the screen the user is viewing. Messaging should be consistent throughout the application and indeed across the full service.

Help — there are three types of ‘Help’ — tutorial help, a hierarchical help system, or context-sensitive help. All three should be used if the platform supports them. The first could be, for example, a guided tour around the application explaining the fundamental user interactions. Hierarchical help has full search methods to find out any information on the application. Context sensitive help, is generally available for each screen, and will only offer help specific to the information on the screen. This help is often activated through a dedicated ‘Help’ key. ‘Help’ should be consistent throughout the application and indeed across the full service.

- Navigational consistency

Payment — the way in which a user may pay for a service or purchase a product must be consistent throughout all the applications. The user will require the mechanism to enter their payment card number, and expiry date, before then agreeing to the transaction. Security precautions may be required, in a form of a PIN number.

Exit — the user must be able to exit an application and service in a consistent way. The method of exit and the information provided must be consistent, from wherever they leave the service.

Time outs — time outs are used throughout applications. Screens may ‘time out’ and return to a previous point in the application, play a video after a certain time of inactivity, or automatically exit the user from the service, if the time elapsed for inactivity is exceeded. The types of time outs within the service must be documented. The service time out must be consistent for all applications.

3.2.2 Media visualisation

It is essential that early visualisation of the screen designs are viewed on the target display as soon as possible. The target hardware can vary greatly from the development environment. For example, most graphic development is done using a computer, but the final graphics might be displayed on an LCD or TV screen. These displays can vary the contrast, brightness and luminance of the graphic image, as well as allowing colours to ‘bleed’ or flicker.

3.3 Functional elements of the process model

3.3.1 Requirements capture

General requirements capture will have been carried out during service design. These requirements will document all the aspirations of the typical potential customers. Not all of these will be realised on the platform.

Requirements capture during application design, focuses more on the possible, and covers in detail the navigation and contents to be designed into the application. Through these detailed workshops with the customer, information can be gathered and detailed into a requirements document. The document will contain:

- a high-level application definition, describing the application,
- detailed information on the navigation, and the number of high-level options or areas within the application,
- information on the scope of content to be presented in the application,
- tasks to be performed — purpose of task, user input to task, task outputs, task frequency, task goal, task dependencies, task duration, task flexibility.

Even, at this stage, some aspirations will not necessarily be met by the hardware or software. Changes can be requested to enhance the functionality of the platform to accommodate the customer’s requirements.

These requirements are then taken a step further and developed into storyboards.

3.3.2 Storyboarding

Storyboarding is a very useful technique for visualisation of the structure and navigation of a multimedia application, following the requirements capture stage.

The technique was first used in animation, but is now widely adopted for multimedia application design.

Storyboards allow the interaction and graphic designer to lay out and design an application in visual terms. Quick sketches can be linked together, allowing the application to grow, and for different navigation flows and application ideas to be visualised.

Storyboarding is successful because at an early stage it is very flexible, allowing the designer to sketch out many different ideas.

Storyboarding is used throughout the early stages of application design. The three types are:

- concept storyboards,
- visual storyboards,
- technical storyboards.

Concept storyboards can be as simple as rough sketches on 'post-it' notes on a large wall. The interaction designer can sketch out the screen images on individual notes. This technique allows the designer a chance to explore the navigation flow of an application, being completely flexible in moving the 'post-it' notes or 'screens' around. This process can be done, with little drawing experience. Although the sketches need to be recognisable, text labels can be used to remind the designer of the screen image. At this stage of design, the storyboard is an excellent brainstorming tool.

This whole process allows the designer to map out the navigation hierarchy of the application and the individual tasks executed within the application, from the start of the application right through to the end product presentation, i.e. define the application structure.

Once a certain design has been agreed, using a simple flow chart documentation, the concepts and application tasks can be fleshed out further in the form of visual storyboards.

Visual storyboards are also created early on in the process of application design. Application flow or navigation will normally have been agreed by this stage, but sometimes it is quite usual to visualise alternate design ideas.

Usually a 'slice' through the application is visualised. A little more care is taken with the storyboard sketches. At this stage thought is also given to application consistency and the possible layout of the screens, e.g. where the titling or the navigation buttons might be placed. Consideration is given to the exact interaction the user would make on screen.

This visualisation can be done on paper or computer. Multimedia developers with Apple Macintosh might consider using HyperCard. Individual cards can represent individual screens.

Concept storyboards represent the hierarchical and sequential nature of an application. Though a multimedia application may have a hierarchical structure, generally there is more than one route through the application. Applications tend to have numerous dynamic links, allowing the user to jump from one place in the hierarchy to another. For example, browsing through places of interest in

a tourist resort, a user might want to book tickets to a theatre. This might take the user from the 'What's on' strand to the 'Reservations' strand of the application. This linking must be recorded in visual storyboards.

Visual storyboards are often supported by early screen prototypes. These aid visualisation for the customer who can, using the visual storyboards, map them to screen prototypes to imagine the end application. Where possible, the prototype screen designs should be viewed on the delivery platform, e.g. TV screen, LCD display or a computer screen. This ensures that the customer has full understanding and visibility of the application at a very early stage of design.

This level of storyboarding still allows a great level of flexibility and is very cost effective.

Following the visual storyboard stage, these ideas are further tested by turning the storyboards into a concept demonstrator.

Technical storyboards will only be briefly described. This stage is usually completed following the sign off of the design brief by the customer.

These storyboards will be produced for every single navigational screen and provide template information for large amounts of product data information. This storyboard gives full detail of every possible navigation to and from the screen, provides complete references to all graphics, sound files, or video material that make up a complete screen. As well as references, exact screen co-ordinates are also supplied. This allows the 'application' builder to take all source material and build the application with ease.

3.3.3 Concept demonstrator

A concept demonstrator is normally a 'stand-alone' demonstrator, that mimics the target platform as much as possible. Graphic designers can work with the visual storyboards, to provide full-screen graphics for the prototyper to build into an application.

The proposed navigational structure is put together, using a commercially available authoring package, and populated by the screen graphics. Two authoring packages that are frequently used to build demonstrators are 'Authorware' and 'Supercard'. Both are very easy to use, and a prototyper can quickly build a demonstrator.

This demonstrator can fully represent the different application strands. Only a subset of product media content is required for the demonstrator to be effective. A fully populated demonstrator ensures that it can be thoroughly user tested. A demonstrator that only has one 'route' through it may not fully reflect the application design, it

means that only certain aspects of the application design can be tested. In certain circumstances this may be acceptable.

Having established a navigational model during the service design stage, user interface consistency and branding will be implemented into the demonstrator. The effectiveness of this model will be evaluated by subsequent user testing.

- User interface consistency

Screen layout is a key to achieving UI consistency. It ensures that the user can navigate around the application with ease, without losing their sense of place in the application hierarchy. Consistency is achieved by designing templates. These template screens would have specific layouts, specifying text areas for titling, body text, possible pricing information, or image positioning. Using these templates for certain screens promotes a consistent 'look'. For example, a product detail screen would have a title of the product at the top, a description on the left, a product image on the right, and possibly a price below the product description text.

UI consistency, however, does more than suggest layout consistency. It refers to consistent use of colour, text sizes and fonts. Therefore, a product detail screen, would not only specify certain areas for text or graphics, but would also reference font type, font size, font colour and number of colours used for the graphic. All these interface details are reflected in the demonstrator.

- Branding

Branding is essential to communicate the brand values of the application provider to the users. The application provider may be BT or indeed an information provider. The brand values expressed by BT, are those which it wishes to ascribe to the application. This is achieved through implementation of the BT branding system, thus differentiating our applications from other information providers and our service from those of our competitors. BT's brand values and branding system would be fully defined in the application styleguide.

If the service is to take on a new branding approach then this needs to be designed and implemented using a new branding system. This can be evaluated during user testing.

3.3.4 User testing

User testing takes place at two different levels — expert evaluation and end user testing.

Expert evaluation is carried out usually by a human factors expert. The evaluator assumes the role of a naive end user, and will have had no previous exposure to the application. This evaluation session will predict which areas of the application may cause the end user difficulty.

Following the initial evaluation, the team at this stage may recommend that changes are made before the demonstrator is fully end user tested.

End-user testing can only take place after planning for the tests has been completed. The evaluation team will decide on different scenarios which the end user will have to carry out to fully test the functionality of different strands of the application, e.g. 'During your flight you wish to make a telephone call to number 123456, using credit card 654321.'

Planning also includes recruitment of the end users. This is normally done through a recruitment agency. The user testing environment will, where possible, mimic the environment of the users of the service. Sessions of 8 to 10 end users are normally set up at any one time. Normally, a democratic representative sample of end users are used.

The human factors facilitator will ask users to perform a certain task. The users are asked to talk aloud, so that their thoughts and decisions can be recorded. This is followed by group discussions on the usability of the application in delivery of the task. Discussions can also be targeted at the naming of particular categories, or the implementation branding system.

The full session is recorded to audio tape and video tape. Following the user testing sessions, the evaluation team will study these, and write a document outlining the good and bad areas within the demonstrator, and, where appropriate, will make recommendations for change.

These changes will be discussed with the customer. If any of the changes are agreed, these are reflected back into the visual storyboards and, the concept demonstrator. Further user testing may be required, if significant changes have had to be made.

3.3.5 Design brief

The design brief is a written high-level definition of the application to be designed. It takes into account all the requirements of the customer, the functional specification of the platform, and consistency issues raised through the styleguide. The visual storyboards and demonstrator support this document.

The design brief, on agreement, is signed by the customer, and is the absolute basis for the development of the application. The relevant technical experts use it to

produce the detailed technical storyboards, data models, logical designs and test cases for verification, validation and testing. It is the brief for multimedia designers to produce the navigational content and product content for the application.

The design brief contains information on:

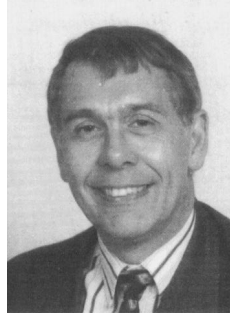
- application overview — detailing the navigation hierarchy of the application,
- application functions — detailing the full depth of navigation in the application, normally through one strand of the application, entering and exiting the application, getting help, error and information messaging, time outs,
- application contents — detailing fixed content, maintainable content and how regularly content will have to change,
- application branding — detailing the application styling and branding,
- usage context — detailing the user population,
- constraints — detailing constraints and restrictions on the application,
- deliverables — detailing deliverables,
- key dates and time-scales — detailing scheduled time-scales,
- usage statistics — detailing the usage statistics required by the service provider and information providers,
- acceptance criteria — detailing criteria for application acceptance, e.g. reviews and checkpoint meetings.

4. Conclusions

The service design process, as embodied in Fig 1, the 'service design pyramid', is fundamental to the successful design of a multimedia service and must be complete before the commencement of application development.

There must be sufficient documented requirements from potential information providers for each of the business strands within a service to enable the hardware and software platform to be selected.

The styleguide document should be substantially complete before starting application development and must document service values and give precise details of the branding system. The consequences of not doing so, could result in major rework of application designs with severe implications on time and budget.



Gary Smith joined BT in 1970 after completing a PhD at Leeds University. He has five years' experience in multimedia service creation and currently leads a multimedia interface design group.

He is one of the founder creators of BT's interactive multimedia services for airlines. His team is now responsible for service creation and delivery for British Airways, Singapore Airlines and Lufthansa.

He is currently working on 'BT Interactive TV'. As well as overall responsibility for UI delivery, he personally manages the 'Top Level' and 'Adland' applications. He is also managing application creation for the Kiosk World programme.



Wendy Jones graduated from Loughborough University in 1990 with an honours degree in Ergonomics. After gaining an MSc in Information Technology, specialising in telecommunications, at the same institution in 1991, she joined BT. During her time in Customer Premises Technology Division, she helped develop numerous user interfaces (UI) for future CPE and early multimedia concept work. She joined Human Factors in 1993 and continued in multimedia UI design, developing applications for airline interactive services (ALIS). She is now with the Multimedia Applications Unit, continuing

with the ALIS programme. She has recently been responsible for developing and writing a styleguide to support UI design consistency for 'BT Interactive TV' and continues to support the application development teams by providing interaction design consultancy. Currently, she is project managing the UI work package for Kiosk World.